

09 — Glossary and Annotated References

What you'll learn: quick definitions of the recurring terms, and an annotated, linked bibliography of every source with a one-line "read this for X."

Prereqs: none (reference doc).

Why it matters here: a single place to look things up and to find the primary sources.

Glossary

- **Gaussian Process (GP)** — a distribution over functions; any finite set of outputs is jointly

Gaussian, defined by a mean function and a kernel. doc 01

- **Kernel / covariance function** $k(x, x')$ — encodes how correlated two outputs are; sets

smoothness, amplitude, reach. doc 02

- **Matérn kernel** — kernel family with tunable smoothness ν ; $\nu=3/2$ (once-differentiable) is our

default. Exactly representable as an SDE/Kalman model and the SPDE Laplacian model. 02, 05, 06

- **Lengthscale** ℓ — input distance over which outputs stay correlated; the main smoothness knob.
- **Signal variance** σ^2 / **noise variance** σ_n^2 — amplitude of the function / of observation noise.
- **ARD** — automatic relevance determination; per-input lengthscales that let the model ignore

irrelevant inputs. 02

- **Multitask / multi-output GP (LMC/ICM)** — joint GP over correlated outputs via a coregionalization

matrix B ; used for correlated (v_x, v_y, v_z) . 02 §8

- **Marginal likelihood** — probability of the data under the GP prior; maximized to learn hyperparameters; auto-balances fit vs complexity. 01 §5

- **Predictive mean / covariance** — the GP's outputs at a query point; the covariance is what we feed

the EKF as adaptive R. 03 §6

- **Cholesky factorization** — stable LL^T decomposition used to solve GP equations; the $O(n^3)$ step.

03

- **Jitter** — tiny diagonal added for numerical stability. 03 §5
- **Error-state (indirect) EKF** — filters the small error between truth and an IMU-propagated nominal

state; this repo's filter. 04 §1

- **Adaptive measurement noise** R_k — using a per-measurement, GP-predicted covariance instead of a

fixed R ; the core fusion idea. 04 §2-3

- **Mahalanobis distance** — covariance-aware distance for outlier gating. 04 §2
- **DVL (Doppler Velocity Log)** — acoustic sensor giving body-frame velocity (here, resolved 3-D);

loses **bottom-lock** over soft/steep terrain → dropouts.

- **Dead-reckoning** — integrating velocity/acceleration to get position; drifts without absolute fixes.

- **SDE (stochastic differential equation)** — differential equation driven by white noise; a temporal

GP's state-space form. 05

- **State-space GP / Kalman duality** — Markovian temporal GP $\equiv$ Kalman filter/RTS smoother; $O(n)$.

05

- **Precision matrix** $\mathbf{Q} = \mathbf{\Sigma}^{-1}$ — inverse covariance; sparse for Markov processes (block-tridiagonal

in time, neighbour-sparse in space). 05 §3, 06 §3

- **SPDE (stochastic PDE)** — PDE with random forcing; $(\kappa^2 - \Delta)^{\alpha/2} X = ?$ has a Matérn solution — the

PDE $\leftrightarrow$ GP duality. 06

- **Laplacian** Δ — the second-order operator $\Sigma \partial^2 / \partial s^2$; the "second-order PDE" in the duality.
- **GMRF (Gaussian Markov Random Field)** — Gaussian field with sparse precision; the FEM solution of

the SPDE. 06

- **FEM (finite element method)** — mesh-and-basis discretization that turns the SPDE into a GMRF.
- **Advection / diffusion** — transport by a velocity field / spreading by mixing; the physics encoded

in the current-field SPDE. 07 §2

- **Dirichlet / Neumann / Robin boundary conditions** — fix value / normal-derivative / a mix on the

domain boundary. 07 §3

- **Inducing points (FITC/SVGP)** — $m \ll n$ summary points for approximate large-scale GPs.

08 §2d

- **GPyTorch** — PyTorch-based GP library (autodiff hyperparameters, multitask kernels, SVGP); our

demo-build tool.

Annotated references

GPs for underwater / AUV navigation

- Cohen & Klein (2025), Gaussian Process Regression for Improved Underwater Navigation —

arXiv:2502.16510. Read for: the exact pattern we follow — MOGPR on DVL feeding adaptive R_k into

an error-state EKF; $\sim 20\text{--}42\%$ gains; composite ARD-SE+Matérn $3/2$ +RQ kernel; real-time cost caveat.

<<https://arxiv.org/abs/2502.16510>>

- **Position Estimation for UVs using UKF with GP Prediction (GP-UKF)** — Read for: learning the

motion/observation model with a GP inside a UKF; graceful degradation.

- Ko & Fox, GP-BayesFilters — Read for: the foundational GP-in-Bayes-filter framework.

Continuous-time trajectory estimation / state-space GPs

- Barfoot, Tong, Anderson (2014), *Batch Continuous-Time Trajectory Estimation as Exactly Sparse GP

Regression* — arXiv:1412.0630. Read for: the block-tridiagonal precision, white-noise-on-

acceleration prior, $O(n)$ trajectory estimation, query/interpolation.

<<https://arxiv.org/abs/1412.0630>>

- Särkkä, Solin, Hartikainen (2013), *Spatiotemporal learning via infinite-dimensional Bayesian

filtering and smoothing* — Read for: the kernel $\leftrightarrow$ SDE $\leftrightarrow$ Kalman equivalence and $O(n)$ temporal GPs.

- Särkkä & Solin, Applied Stochastic Differential Equations (2019, free) — Read for: the math

of turning kernels into SDEs. <<https://users.aalto.fi/~ssarkka/>>

- Hartikainen & Särkkä (2010), Kalman filtering and smoothing solutions to temporal GP regression.*

The SPDE / PDE duality

- Lindgren, Rue, Lindström (2011), *An explicit link between Gaussian fields and Gaussian Markov

random fields: the SPDE approach*, JRSS-B 73(4) — Read for: the core duality and the FEM $\rightarrow$ GMRF

sparse-precision construction.

<<https://portal.research.lu.se/en/publications/an-explicit-link-between-gaussian-fields-and-gaussian-markov-rand/>>

- **Whittle (1963)** — Read for: the original Matérn $\leftrightarrow$ SPDE observation.
- Bakka et al. (2018), Spatial modelling with INLA: a review — Read for: an accessible tour of

SPDE/GMRF modelling and the R-INLA tooling.

• Lindgren, Bolin, Rue (2022), The SPDE approach ... 10 years and still running, Spatial Statistics — Read for: the modern state of the field.

Physics-informed GPs and boundary conditions

- Clarotto, Allard, Romary, Desassis (2022/2024), *The SPDE approach for spatio-temporal datasets

with advection and diffusion*, Spatial Statistics — arXiv:2208.14015. Read for: the advection-diffusion SPDE (currents), nonseparability, streamline-diffusion stabilization.

<<https://arxiv.org/abs/2208.14015>>

- Gulian / Tan et al. (2024), *Boundary-constrained Gaussian processes for robust physics-informed

ML of linear PDEs*, JMLR 25 — Read for: exact Dirichlet/Neumann/Robin-constrained mean+kernel

design. <<https://www.jmlr.org/papers/v25/23-1508.html>>

• Jidling, Wahlström, Wills, Schön, Linearly constrained Gaussian processes — Read for: divergence-free / curl-free vector-field kernels (incompressible currents).

- **Physics-informed, boundary-constrained GPR for fluid flow fields (2025), arXiv:2507.17582** —

Read for: a worked fluid-reconstruction example with incompressibility + boundaries.

Scalable / sparse GPs (edge compute)

- Rasmussen & Williams (2006), Gaussian Processes for Machine Learning (free) — Read for:

everything foundational (chs. 2-5). <<https://gaussianprocess.org/gpml/>>

- **Quiñonero-Candela & Rasmussen (2005)** — Read for: a unifying view of sparse approximate GPs

(SoR/FITC/DTC).

- Hensman, Fusi, Lawrence (2013), Gaussian Processes for Big Data — Read for: SVGP / variational

sparse GPs.

- Wilson & Nickisch (2015), KISS-GP — Read for: structured-kernel near-linear inference on grids.
- **GPyTorch (Gardner et al., 2018) + docs** — Read for: the library we'll use; multitask GPs, SVGP,

exact GP tutorials. <<https://gpytorch.ai/>>

Further reading

- Duvenaud, The Kernel Cookbook — kernel-composition intuition.
<<https://www.cs.toronto.edu/~duvenaud/cookbook/>>
- A Visual Exploration of Gaussian Processes (distill.pub) — interactive intuition.
- Back to the overview or on to the

implementation plan.